Group and solo flights

Does flying in a group change how a paraglider moves?

Matteo Di Sante

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Abstract

Do paragliders move differently when they fly in a group? Group time differs from solo flight in many ways, but so do the pilots who fly in groups, the places and hours where groups form, and the moments pilots choose to fly together. We measure the motion by the local exponent $H_{\mathrm{loc}}(\tau)$ of the time-averaged mean squared displacement and by statistics of the climbs and glides, and take each of the other causes away with one test: contrasts within strata of site, day and hour and within the same flight; solo flights reweighted to the zones and hours of the groups; the same flight before, during and after a group. Over a quarter of an hour a pilot in a group progresses straighter. $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ is $+0.037$ against the same flight alone and $+0.043$ against the same flights alone at the same site, day and hour, while who flies in groups accounts for $+0.003$. The zones and hours where groups fly explain a fifth to a quarter of the raw gap, and on the same flight the straighter progress starts and stops with the group. Within one glide or climb the progress is no straighter. Glides are faster and flatter in a group, both within strata and on the same flight before and after; pilots also climb more, but on the same flight the climbs change when the group forms and not when it breaks. Most of this is there with two pilots, and the pilot ahead ends searches in a climb as often as the one behind. The tests fix the pilot, the place, the hour and the moment, but not what brings pilots together: the effect goes with flying together, and whether the group causes it stays open.

1 Introduction

The thesis studies how paraglider pilots move in cross-country flight: how long they climb, how far they glide, how their displacement grows with time. Flying in a group could change all of this. In soaring birds, social information on the position of thermals changes the gliding strategy [2], and in a migrating flock a few individuals find the thermals while the others follow them [1]. The note *Group flights in free flight* (`ideas/group-identification`) labels every instant of every flight of the FFVL archive as solo or in a group. This report asks whether, on the same archive, pilots move differently when they fly in a group.

A raw comparison cannot answer, because group time differs from solo flight in four other ways.

(i) **The pilots.** Those who fly in groups may be more experienced, or fly faster wings.
(ii) **The place.** Groups form on some sites and routes, where any pilot may progress straighter.
(iii) **The day and the hour.** Group time gathers around midday and on some days more than others, and the air of that day and hour acts on every pilot.
(iv) **The moment.** Pilots may join others when the progress is straight anyway, along a convergence line or a ridge, and the straight progress would then come before the group.

Each of these can produce a difference between group and solo time with no effect of the group. The report removes them one at a time. Section 2 fixes the populations and the measures, and Section 3 shows the raw difference. Contrasts within strata of site, day and hour, and within the same flight, separate the pilots, the day and the hour from the group (Section 4); reweighting the solo flights to the zones and hours where groups fly tests the place (Section 5); the same flight before, during and after a group tests the moment (Section 6). What survives the three tests we call the group effect. Section 7 looks inside the group, at its size and at who leads, and Section 8 says what can and cannot be concluded.

2 Data and measures

Which flights, which populations, and which measures of the motion?

Flights and groups. We use 156 391 paraglider and 6 094 hang-glider flights on 6 209 days, the cleaned tracks of the old repository (soaring, tag `legacy-final`). A flight is analysed after its *departure*, the first fix

3 km from the takeoff; before it the pilot counts as neither solo nor in a group. The measures of the motion use loggers of at most 10 s; the phases (climb, search, glide) are Vilpellet's runs and exist on the flights logged every second, about 70% of the paragliders. The groups are those of the note, with $\delta = 2$ min, $\varepsilon_h = 500$ m, $\varepsilon_z = 300$ m, $\theta_{\max} = 45°$, $g = 2$ min and $q = 0.1$, hence contacts of at least $\tau^* = 38.1$ min (its sections 3–6). A contact means shared air within sight; it does not prove interaction.

What the grouping gives. Of the 154 598 paraglider flights with a logger of at most 10 s, 33 501 (21.7%) fly in a group at some point; of the 6 039 hang-glider flights, 418 (6.9%). Group time is 12.8% of all paraglider flight time after the departure, and on average 45% of the time of a group flight. The 36 575 pair intervals last 40 to 96 min (10th to 90th percentile; median 52 min) and form 40 051 configurations in 17 169 episodes. Half of the episodes have two flights, 1% have more than 9, the largest has 53. By the size of the configuration, the paraglider group time is: 2 pilots 64%, 3 pilots 19%, 4 pilots 8%, 5-6 pilots 6%, 7+ pilots 3% (the largest configuration has 27 pilots). In contact, one pilot is behind the other by a median of 20 s (75% within 45 s).

Populations. Three populations of stretches of flight.

- S, *solo flights*: flights never in a group.
- G, *group time*: the stretches of a flight inside a configuration, a fixed set of flights that stays one group over a maximal interval. G_n is the group time in configurations of n pilots; "n pilots" always means the size of the configuration at that instant.
- A, *the same flights alone*: the stretches of the group flights outside any configuration. Nearly every group flight has some (Table 1), so A has nearly the pilots, gliders, loggers and days of G, and serves as its control.

The motion over every lag. The horizontal position $\mathbf{r}(t)$ of a flight is read on a 10-s grid of its clock. A *piece* is the part of a stretch within one segment of the track; it enters only if it lasts at least 1250 s, so that every population keeps the same flights at all 21 lags from 10 to 1000 s. For a flight f and a population, the time-averaged mean squared displacement pools the origins of all its pieces in that population,

$$\overline{\delta_f^2}(\tau) = \frac{1}{N_f(\tau)} \sum_t \left\| \mathbf{r}(t+\tau) - \mathbf{r}(t) \right\|^2, \tag{1}$$

where t runs over the $N_f(\tau)$ origins with t and $t+\tau$ in the same piece, and the ensemble $\mathrm{MSD}(\tau)$ is the mean of $\overline{\delta_f^2}(\tau)$ over the flights, one weight per flight. With $\mathrm{MSD}(\tau) \propto \tau^{2H}$, H is half the slope of $\log \mathrm{MSD}$ against $\log \tau$ over all lags: $H = 1$ for a straight line, $H = 1/2$ for a random walk. $H_{\mathrm{loc}}(\tau)$ is the same slope over the lags within ± 0.25 decades of τ (Appendix A). We read it at two scales: $\tau = 100$ s, within one climb or one glide, and $\tau = 1000$ s (lags 580–1000 s), over one to two climb–glide cycles.

Climbs and glides. Each phase run goes whole to the population of the stretch that holds its middle, and each statistic is a ratio of sums over the runs of a population, so long phases weigh more. Of the twelve statistics measured, five are not explained by the others and are the ones drawn (Appendix B): the share of time in climb; the share of searches that end in a climb; the climb rate, total height gained over total climb time; the glide speed, total straight-line distance over total glide time; and the glide ratio, total distance over total height lost, replaced in stratified contrasts by its inverse, the height lost per kilometre of glide. The tables also give the cross-country speed, the straight-line distance covered in 30 min averaged over every origin, which the five explain.

Strata. A *zone* is a cell of 25 km containing the middle of a piece; a *site-day* holds the flights of one day whose takeoffs lie within 3 km; the *hour* is the local hour of the middle of a piece or a run.

Uncertainty. Bands are 5–95% of bootstrap draws that resample whole dependency blocks, the flights tied by sharing episodes, and recompute the statistic: the blocks of the note for the pooled values of Tables 1 and 4 (1000 draws); blocks that also join the flights of a site-day for every contrast (200 draws, the same for every statistic). Every difference is computed draw by draw and has its own band. The bands measure the sampling of blocks; the choice of q is tested in Appendix C.

What we keep. Three populations, S, G and A; two scales of the motion, $H_{\mathrm{loc}}(100\ \mathrm{s})$ and $H_{\mathrm{loc}}(1000\ \mathrm{s})$; five statistics of climbs and glides; bands from blocks of flights that are not independent.

3 The raw difference

How do group time and solo flights differ before any control?

	solo flights S	group time G	same flights alone A
flights (TA-MSD)	119 129	33 435	29 611
H (10–1000 s)	0.899 [0.898, 0.899]	0.916 [0.915, 0.917]	0.914 [0.913, 0.915]
$H_{\mathrm{loc}}(100\ \mathrm{s})$	0.919 [0.919, 0.919]	0.927 [0.927, 0.928]	0.932 [0.931, 0.932]
$H_{\mathrm{loc}}(1000\ \mathrm{s})$	0.837 [0.837, 0.838]	0.893 [0.891, 0.894]	0.860 [0.859, 0.862]
time in climb (%)	30.8 [30.7, 30.9]	33.0 [32.8, 33.2]	29.1 [28.9, 29.2]
searches ending in a climb (%)	43.3 [43.2, 43.4]	49.5 [49.2, 49.7]	45.9 [45.7, 46.1]
climb rate (m/s)	1.22 [1.22, 1.22]	1.26 [1.25, 1.27]	1.33 [1.32, 1.34]
glide speed (km/h)	30.1 [30.1, 30.2]	33.3 [33.1, 33.5]	32.8 [32.6, 32.9]
glide ratio	10.4 [10.4, 10.4]	12.9 [12.8, 13.0]	11.3 [11.2, 11.4]
cross-country speed (km/h)	17.2 [17.2, 17.3]	20.0 [19.8, 20.3]	20.0 [19.8, 20.1]

Table 1: Pooled statistics of S, G and A (paragliders), with 5–95% bands. Exponents on loggers of at most 10 s, phases on 1 Hz flights.

Group time progresses straighter over long lags than solo flights: $H_{\mathrm{loc}}(1000\ \mathrm{s})$ is 0.893 in group time against 0.837 in solo flights. It also spends more time climbing, ends more searches in a climb, glides faster and glides flatter, with a glide ratio of 12.9 against 10.4 (Table 1; the curves over every lag are in Appendix D). The same flights alone sit between the two at long lags (0.860) and have the highest climb rate. Part of the raw difference may therefore belong to who flies in groups, and to where and when they fly.

What we keep. A difference to explain, largest at long lags and in the glides. The next three sections ask how much of it is the group.

4 Group or pilots? Contrasts within strata

How much of the difference belongs to who flies in groups, and how much to the group, once the day, the site and the hour are fixed?

Contrast. For a statistic X and two populations P and Q, each stratum s that holds both gives $D_s = X_{P,s} - X_{Q,s}$, computed as on the whole archive (ensemble MSD, ratios of sums). The contrast is their weighted mean,

$$\Delta_{P-Q} = \frac{\sum_s w_s D_s}{\sum_s w_s}, \qquad w_s = \frac{n_{P,s}\, n_{Q,s}}{n_{P,s} + n_{Q,s}}, \tag{2}$$

with $n_{P,s}$ and $n_{Q,s}$ the flights of each population in s. The weight is half the harmonic mean of the two counts: a stratum counts as much as its smaller side allows. For the exponents, D_s is the difference of log MSD at every lag, and ΔH and ΔH_{loc} are half the slopes of its weighted mean, over all lags and within ± 0.25 decades.

Five contrasts (Figure 1; values with bands in Table 9, Appendix E):

- G − S pooled: the raw difference;
- G − S within site-day and hour: day, site and hour fixed;
- A − S within site-day and hour: *selection*, how the flights that fly in groups differ from solo flights when they fly alone;
- G − A within the flight, the stratum being one flight: pilot, glider, logger, day and site fixed, the hour free;
- G − A within site-day and hour: day, site and hour fixed, the pilots different.

The last two estimate the group effect and fail in opposite ways. Within the flight, group time falls around midday and the time alone of the same flight earlier and later (section 7 of the note), so the hour differs; within site-day and hour, the hour is the same but the pilots differ. Where the two agree, neither the hour nor the pilot explains the difference.

Results. The straighter progress over 1000 s survives both estimates: $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ is $+0.055$ pooled, $+0.037$ within the flight and $+0.043$ at the same site-day and hour, while selection gives $+0.003$. At 100 s the two estimates disagree in sign, -0.009 within the flight and $+0.002$ at the same site-day and hour: within the flight the hour differs, and fixing it removes the difference (Figure 2, right). The pooled $+0.008$ comes from who flies in groups ($+0.005$) and from the day, site and hour.

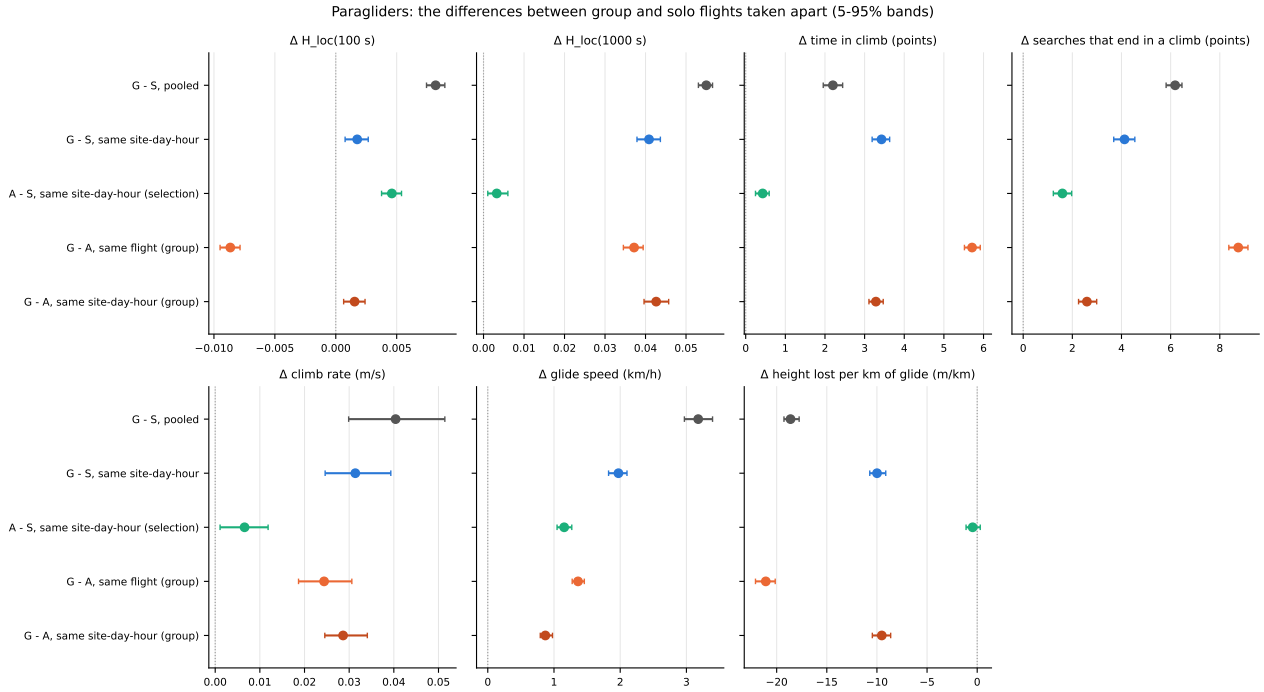


Figure 1: Each statistic (panel) for the five contrasts (rows), with 5–95% bands. Shares of time in points; lower height lost per km is a better glide.

In a group pilots spend more time climbing (+5.71 and +3.28 points), end more searches in a climb (+8.75 and +2.59 points) and climb faster (+0.024 and +0.029 m/s). Pooled, G climbs slower than A (Table 1); within strata it climbs faster. Glides are faster, +1.36 and +0.87 km/h, and flatter, −21.1 and −9.5 m/km; the cross-country speed rises by +0.70 and +1.29 km/h. For the phase statistics the within-flight values are two to three times the site-day-and-hour ones, as expected if the hour adds to them, so the site-day-and-hour values are the conservative estimates. Selection is small for the motion and for the climbs, but not for speed: group pilots glide +1.15 km/h faster even alone, over half of the +1.97 km/h between G and S at the same site-day and hour.

What we keep. With the pilot fixed, or with the day, site and hour fixed, group time progresses straighter over a quarter of an hour, climbs more and glides faster and flatter. Who flies in groups explains little of the straighter progress and over half of the faster glides.

5 Group or place? Solo flights where groups fly

Do groups fly where, and when, any pilot would progress straighter? If so, solo flights compared on the same ground and at the same hours would match group time.

Reweighting. Each piece gets a stratum from its middle, in four stratifications from coarse to fine: zone, zone×month, zone×hour, site-day×hour. The solo flights are then given the mix of the group over the strata (direct standardisation):

$$\mathrm{MSD}_{\mathrm{S|G}}(\tau) = \frac{\sum_s n_{\mathrm{G},s}\,\mathrm{MSD}_{\mathrm{S},s}(\tau)}{\sum_s n_{\mathrm{G},s}}, \tag{3}$$

with $n_{\mathrm{G},s}$ the units of G in stratum s (a flight counts once per stratum it crosses) and $\mathrm{MSD}_{\mathrm{S},s}$ the ensemble MSD of the solo units of s. In the statistic, S then flies where and when G flies. G is taken on the strata it shares with S, so the two curves have the same support; the same is done for A. "As is" is the comparison on the same strata without the reweighting.

Results. The gap opens above $\tau \approx 150$ s (Figure 2): in every stratification ΔH_{loc} between G and S lies between +0.005 and +0.011 from 20 to 140 s, and pooled it reaches +0.055 at 1000 s, above one climb–glide cycle. Reweighting removes little of it. To the zones of G, it lowers $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ by 6% (+0.059 → +0.056) and the fitted ΔH by 15%; with zone×month, by 15% and 23%; with zone×hour, by 19% and 24%. Solo pilots do fly where groups fly: 100% of G has a solo counterpart in its cell. At the same site-day and hour

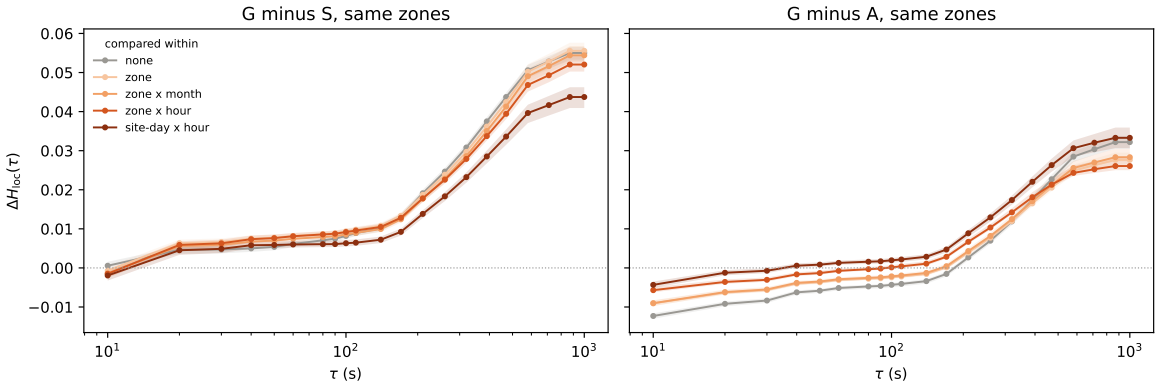


Figure 2: $\Delta H_{\mathrm{loc}}(\tau)$ over every lag: G minus S (left) and G minus A (right), the other population reweighted to the strata of G; one line per stratification, finer strata in darker orange. The curves themselves are in Figure 9.

		fitted H (10–1000 s)			ΔH			$\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$		
compared within	G kept (%)	G	S | G	A | G	G − S as is	G − S reweighted	G − A reweighted	G − S as is	G − S reweighted	G − A reweighted
none	100	0.916	0.899	0.914	+0.017	+0.017 [+0.016, +0.018]	+0.002 [+0.001, +0.003]	+0.055	+0.055 [+0.053, +0.057]	+0.032 [+0.030, +0.034]
zone	100	0.919	0.902	0.916	+0.021	+0.018 [+0.017, +0.019]	+0.003 [+0.003, +0.004]	+0.059	+0.056 [+0.054, +0.058]	+0.028 [+0.026, +0.029]
zone × month	99	0.919	0.902	0.915	+0.022	+0.017 [+0.016, +0.018]	+0.003 [+0.003, +0.004]	+0.064	+0.054 [+0.053, +0.057]	+0.028 [+0.027, +0.031]
zone × hour	99	0.919	0.902	0.913	+0.022	+0.017 [+0.016, +0.018]	+0.005 [+0.005, +0.006]	+0.064	+0.052 [+0.050, +0.054]	+0.026 [+0.025, +0.028]
site-day × hour	53	0.917	0.904	0.913	+0.015	+0.013 [+0.012, +0.014]	+0.008 [+0.007, +0.009]	+0.051	+0.044 [+0.041, +0.046]	+0.033 [+0.031, +0.036]

Table 2: Fitted H and differences by stratification. "Reweighted": the other population reweighted to the strata of G (value, 5–95% band); "as is": the same strata without reweighting. $H_{\mathrm{S|G}}$: H of S reweighted to G. The columns G and "kept" refer to the strata shared with S; on those shared with A they differ slightly.

(53% of G kept), $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ is +0.044 [+0.041, +0.046], 14% below its value as is, and the fitted ΔH is +0.013 [+0.012, +0.014]. Reweighting raises the fitted H of S by a few thousandths (0.899 → 0.904) and never brings it to that of G (0.917).

Against the same flights alone, $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ is +0.032 pooled and +0.028, +0.026 and +0.033 in zone, zone×hour and site-day×hour strata. At the same site-day and hour these values differ from the contrasts of Section 4 on the same strata (+0.033 here against +0.043 there) only through the weights, $n_{\mathrm{G},s}$ here and the half harmonic mean there: the difference varies between strata, and the group effect at the same site-day and hour lies between the two.

Limits. Cells are 25 km, placed by the middle of a piece; the terrain inside a cell is not controlled, and a group may choose better air within it, which is then part of what we call the group effect. Paragliders only.

What we keep. The zones, months and hours where groups fly explain a fifth to a quarter of the gap with solo flights; the rest stays at the same site, day and hour.

6 Group or moment? The same flight before, during and after

Do pilots fly together in the moments when the progress is straight anyway? If so, the same flight would already be straight just before the group forms, and still just after it breaks.

Windows. Every stretch of group time is an *event*, with up to four windows of length $W = 50$ min on the clock of the same flight: *start* and *end*, the first and last W of the group, when it lasts at least W; *before* and *after*, the last W of the solo stretch that precedes the group and the first W of the one that follows it, when that solo stretch lasts at least W, whatever the length of the group. A window of 50 min holds about 200 origins at $\tau = 1000$ s. For two windows the ensemble MSD is taken on the events that have both, and their difference is read on H_{loc} and on the phase statistics. Entering is start minus before; leaving is end minus after, group minus alone in both cases. The two readings predict different things:

- if the straighter progress belongs to the group, it appears on entering and on leaving, and before and after are alike;
- if it belongs to the moment, before is already high, and the changes are those of any two consecutive windows.

Controls. Two consecutive windows with no change of state, drawn in the solo time of the group flights (A) and in solo flights (S), measure how much a flight drifts from one window to the next, later minus earlier. Entering has the same time order as the controls; leaving has the opposite one, so on leaving the drift enters with the opposite sign.

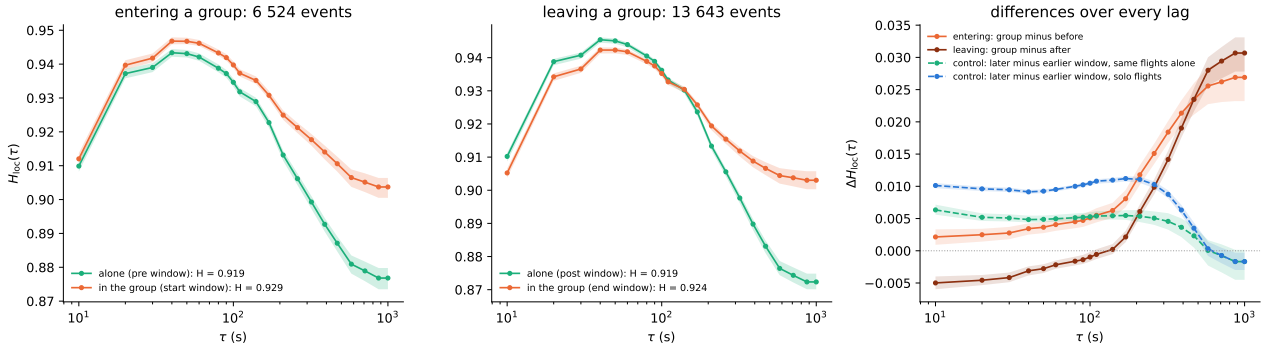


Figure 3: $H_{\mathrm{loc}}(\tau)$ of the same flights alone (green) and in the group (orange), entering (left) and leaving (middle), 5–95% bands. Right: the differences, group minus alone, with the two controls (later minus earlier window; dashed); for leaving, the drift to compare is the control with the opposite sign.

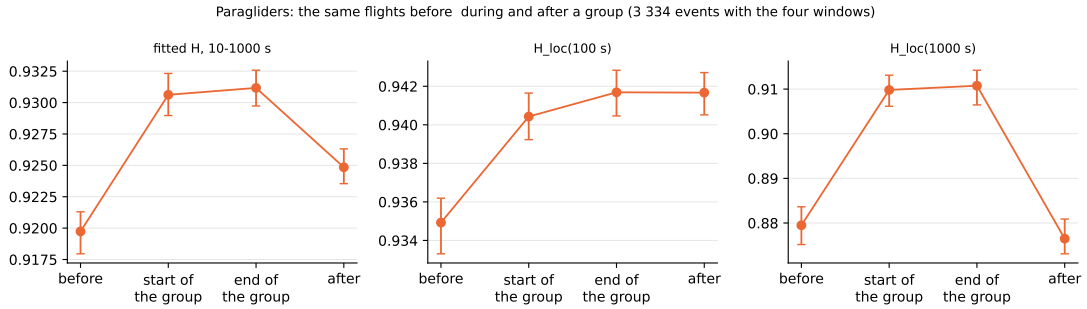


Figure 4: The four windows on the events that have all of them. Start and end overlap when the group lasts less than $2W$.

comparison	events	ΔH	$\Delta H_{\mathrm{loc}}(100)$	$\Delta H_{\mathrm{loc}}(1000)$
entering: start − before	6 524	+0.010 [+0.009, +0.011]	+0.005 [+0.004, +0.006]	+0.027 [+0.023, +0.031]
leaving: end − after	13 643	+0.005 [+0.004, +0.006]	−0.001 [−0.002, −0.000]	+0.031 [+0.028, +0.033]
after − before	6 320	+0.004 [+0.003, +0.005]	+0.006 [+0.005, +0.007]	−0.003 [−0.007, +0.001]
control A: later − earlier	16 998	+0.005 [+0.004, +0.005]	+0.005 [+0.005, +0.006]	−0.002 [−0.005, +0.000]
control S: later − earlier	69 648	+0.009 [+0.008, +0.009]	+0.010 [+0.010, +0.011]	−0.002 [−0.003, −0.000]

Table 3: Differences of the fitted H and of H_{loc} at 100 and 1000 s (value, 5–95% band) between windows, on the events with both windows. Controls: two consecutive windows with no change of state, later minus earlier, in the solo time of the group flights (A) and in solo flights (S).

Results at 1000 s. The first prediction holds. $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ is +0.027 [+0.023, +0.031] entering and +0.031 [+0.028, +0.033] leaving, against drifts of −0.002 and −0.002 in the controls, and after minus before is −0.003 [−0.007, +0.001]. On the 3 334 events with all four windows, $H_{\mathrm{loc}}(1000\ \mathrm{s})$ is 0.879 before, 0.910 and 0.911 in the group and 0.876 after (Figure 4): a step close to the within-flight estimate of Section 4 (+0.037).

Results at short lags and on the phases. Entering, $H_{\mathrm{loc}}(100\ \mathrm{s})$ and the fitted H change by +0.005 and +0.010, about the drift of the controls (+0.005 and +0.010; +0.005 and +0.009). Leaving, they change by

-0.001 and $+0.005$, which exceed the drift taken with the opposite sign: a small asymmetry at short lags, which we do not interpret. Glides are faster and flatter in the group in both directions: glide speed $+0.92$ km/h entering and $+0.52$ leaving, against drifts of $+0.07$ and $+0.04$ km/h; height lost per km -2.3 entering and -10.9 leaving, against drifts of $+3.3$ and $+4.9$ m/km, so that once each drift is taken with its sign the glide flattens by similar amounts both ways (Table 10, Appendix E). The climb statistics rise on entering beyond the drift and do not fall on leaving beyond it.

Limits. The windows fix the pilot, the glider, the day and the slow drift of the flight, not what brings pilots together: a convergence line, a ridge or a common route may last as long as the group and straighten the progress by itself, and two pilots must move alike to stay within 500 m for long. The events are long groups with solo flight beside them (6 524 entries, 13 643 exits). With windows of 21 min the step at 1000 s is not seen (-0.003 entering, $+0.008$ leaving, against drifts of -0.011 and -0.007): such a window holds fewer than thirty origins at 1000 s, so the test at that lag needs windows of 50 min (Appendix F).

What we keep. With windows of 50 min, on the same flight the straighter progress over a quarter of an hour starts with the group and stops with it, and the glides change with it both ways; the moment of joining does not explain them. The climbs change when the group forms and not when it breaks, so for them the moment remains a possible cause.

7 Inside the group: size and leader

Does the effect grow with the number of pilots, and does the pilot ahead fly differently from the one behind?

7.1 Size of the configuration

The group time is cut where the configuration changes, so every stretch has a fixed number of pilots n (2, 3, 4, 5–6, 7+), and a flight can contribute to several sizes. Two contrasts of Eq. (2) follow the size: $G_n - A$ within the flight, and $G_n - G_2$ at the same site-day and hour, which asks whether a group does more than a pair.

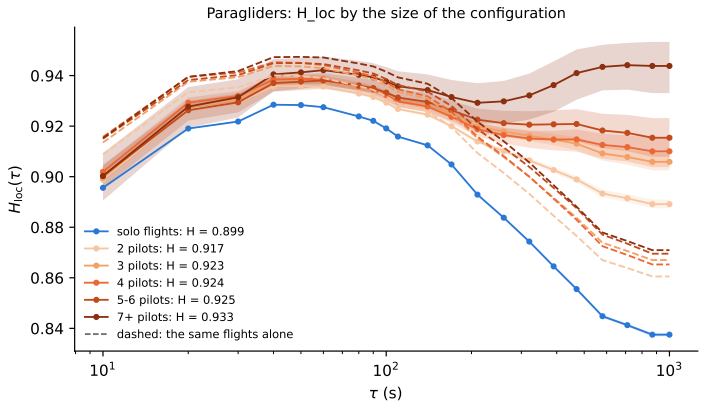


Figure 5: $H_{loc}(\tau)$ over every lag by the size of the configuration (orange, darker with more pilots), S (blue) and, dashed, the same flights alone for each size.

		H		$H_{loc}(1000\ s)$						
pilots	flights	G_n	alone	G_n	alone	time in climb (%)	searches to climb (%)	climb rate (m/s)	glide speed (km/h)	glide ratio
solo	119 129	0.899		0.837		30.8	43.3	1.22	30.1	10.4
2	26 411	0.917	0.914	0.889	0.861	32.7	49.0	1.27	33.2	12.5
3	9 480	0.923	0.919	0.906	0.867	33.6	49.9	1.26	33.5	13.1
4	4 073	0.924	0.920	0.910	0.865	33.6	51.0	1.25	33.5	13.7
5–6	2 686	0.925	0.922	0.915	0.870	33.1	50.1	1.23	33.1	14.1
7+	1 142	0.933	0.924	0.944	0.871	34.6	51.7	1.20	34.6	15.9

Table 4: Paragliders, pooled by the size of the configuration ("alone": the same flights alone).

Pooled, $H_{loc}(1000\ s)$ rises with n, from 0.889 with 2 pilots to 0.944 with 7+, while the same flights alone stay between 0.861 and 0.871 (Figure 5, Table 4). Within the flight the effect also grows, from $+0.026$ with 2 pilots to $+0.087$ with 7+ (Figure 6); there the day and the site are fixed, so the growth comes from the

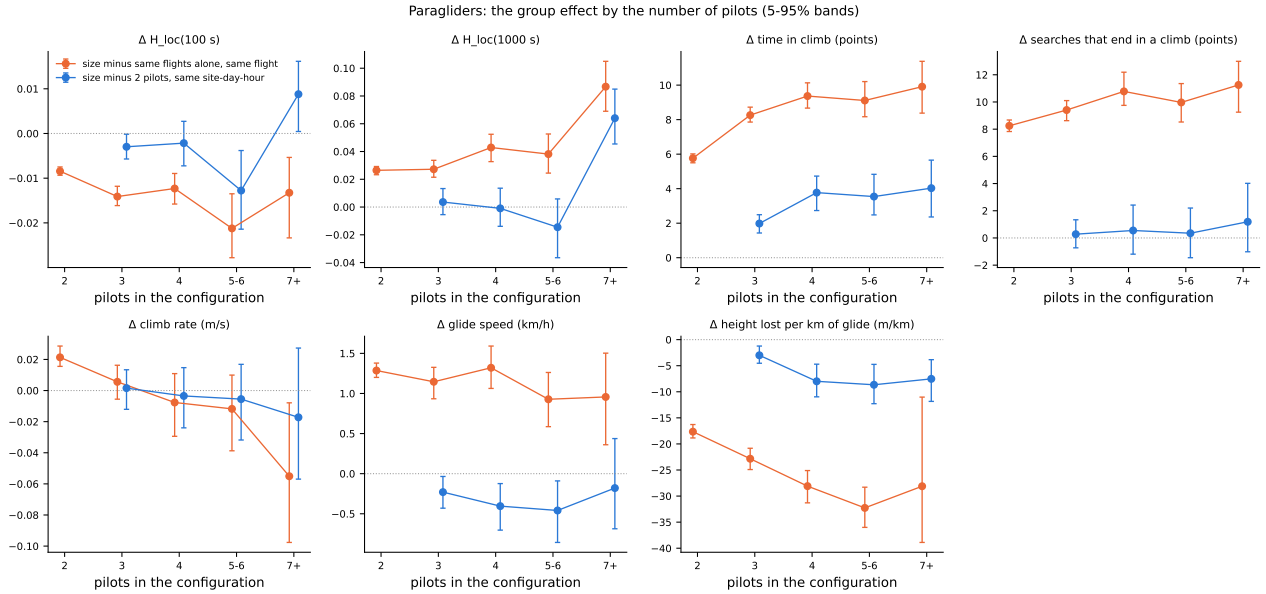


Figure 6: Orange: $G_n - A$ within the flight. Blue: $G_n - G_2$ at the same site-day and hour. Values with bands in Table 11.

hour or from the flights that join large groups. At the same site-day and hour it vanishes up to six pilots: configurations of 3 to 6 pilots differ from pairs by -0.015 to $+0.004$, with bands across zero. Only 7+ stands apart ($+0.064$ [$+0.045$, $+0.085$], cross-country speed $+7.0$ km/h), a rare configuration, 3% of the group time. These may be competition gaggles flying a set task; the archive does not flag competitions, so this stays uncontrolled. Time in climb and the flatter glide do grow with n at the same place and hour, from $+2.0$ to $+4.0$ points and from -3 to -9 m/km. On six single site-days the differences between days exceed those between sizes (Appendix G).

7.2 Leader and follower

In configurations of two pilots, the signed median delay ℓ_{med} of the pair interval (Appendix B of the note) says who passes first: the leader, when the other passes at least 15 s later; otherwise the two fly abreast (35% of the pairs). Leader and follower are compared within the same configuration, the configuration being the stratum of Eq. (2) (Appendix H). The pilot ahead climbs more and a little faster and glides faster (time in climb $+0.91$ [$+0.57$, $+1.25$] points, climb rate $+0.021$ [$+0.014$, $+0.028$] m/s, glide speed $+0.86$ [$+0.74$, $+0.99$] km/h), and glides as flat ($+1.6$ [-0.9, $+4.5$] m/km). The share of searches that end in a climb does not differ (-0.01 [-1.03, $+0.83$] points): on this measure the follower shows no sign of finding thermals through the leader. How groups form and break and whether the same pilots lead are in section 7 of the note.

What we keep. Most of the effect is there with two pilots. Beyond a pair, at the same site-day and hour, size adds climbing and flatter glides but no straighter progress, except in the rare configurations of 7 or more. The leader is a little faster, and ends searches in a climb as often as the follower.

8 What can and cannot be concluded

The raw difference between group time and solo flights is real, and its long-lag part survives the three tests (Table 5). With the pilot fixed, or with the day, site and hour fixed, a group progresses straighter over a quarter of an hour (Section 4); the zones and hours where groups fly explain a fifth to a quarter of the raw gap (Section 5); on the same flight the straighter progress starts and stops with the group (Section 6). At 100 s, within one glide or climb, the progress is no straighter: the difference appears only at lags of one to two climb–glide cycles. Faster and flatter glides pass the contrasts and the before-and-after test; the climbs pass the contrasts but change only when the group forms. Two pilots carry most of it (Section 7).

test	comparison	$\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$	band
raw	G − S pooled	+0.055	[+0.053, +0.057]
pilots, day, site, hour	G − S same site-day and hour	+0.041	[+0.038, +0.044]
	A − S same site-day and hour (selection)	+0.003	[+0.001, +0.006]
	G − A same flight	+0.037	[+0.035, +0.039]
	G − A same site-day and hour	+0.043	[+0.040, +0.046]
place	G − S reweighted, site-day×hour	+0.044	[+0.041, +0.046]
	G − A reweighted, site-day×hour	+0.033	[+0.031, +0.036]
moment	entering, start − before	+0.027	[+0.023, +0.031]
	leaving, end − after	+0.031	[+0.028, +0.033]
	after − before	−0.003	[−0.007, +0.001]
	drift, solo time of group flights (A)	−0.002	[−0.005, +0.000]
	drift, solo flights (S)	−0.002	[−0.003, −0.000]

Table 5: Every estimate of the long-lag difference, from the raw one to the three tests. The drifts are later minus earlier windows; for leaving they enter with the opposite sign.

Supported

- the progress over a quarter of an hour is straighter in a group, with the pilot fixed (+0.037) and with the day, site and hour fixed (+0.043);
- no change of $H_{\mathrm{loc}}(100\ \mathrm{s})$, within one glide or climb;
- on the same flight the straighter progress starts and stops with the group, and before and after are alike;
- faster and flatter glides, within strata and on the same flight both ways;
- more time climbing, more searches that end in a climb and a higher climb rate within strata (pooled, G climbed slower than A); on the same flight these rise when the group forms but do not fall when it breaks;
- selection explains little of the straighter progress and over half of the faster glides;
- the leader is a little faster and ends searches in a climb as often as the follower.

Not supported

- that the group *causes* the effect: the tests fix pilot, day, site, hour, zone and the drift of the flight, but not what brings pilots together (a line, a ridge, a route) nor the coordination needed to stay in contact;
- that larger groups progress straighter than pairs at the same site-day and hour, up to six pilots.

Caveats

- the glide statistics are partly fixed by definition, since headings must be aligned in glide for a contact;
- configurations of 7 or more may be competition gaggles, which the archive does not flag;
- the leader is assigned from ℓ_{med}, which is uncertain in thermals (note, Appendix B).

Open. The phases exist on 1 Hz flights only: the same analyses on the tracks of xc-dynamics, with phases on all loggers, and on hang gliders, which have few group flights (Appendix D), remain to be done. The reproduction of the report is in Appendix J, and the checks of the viewer and of the analyses in Appendix I.

References

[1] A. Flack, M. Nagy, W. Fiedler, I. D. Couzin, M. Wikelski. From local collective behavior to global migratory patterns in white storks. *Science* 360(6391):911–914, 2018.

[2] H. J. Williams, A. J. King, O. Duriez, L. Börger, E. L. C. Shepard. Social eavesdropping allows for a more risky gliding strategy by thermal-soaring birds. *Journal of the Royal Society Interface* 15:20180578, 2018.

A How H and H_{loc} are computed

$\mathrm{MSD}(\tau) \propto \tau^{2H}$, so $\log_{10}\mathrm{MSD} = 2H\log_{10}\tau + c$. H is half the slope of the least-squares line over all 21 lags (equal weight per lag). $H_{\mathrm{loc}}(\tau_c)$ is half the slope of the line fitted on the lags with $|\log_{10}\tau - \log_{10}\tau_c| \leq 0.25$ (a factor 1.78 on each side), equal weight; where fewer than 3 lags lie in the window, the 3 nearest are used. It is a windowed regression over about half a decade. ΔH and ΔH_{loc} of two curves are the differences of the fitted values; for a stratified average they are the same regression on the averaged $\Delta\log\mathrm{MSD}$ (linear: slope of a difference = difference of slopes). Bands: the same computation on each bootstrap draw, percentiles lag by lag.

τ (s)	lags used	τ (s)	lags used	τ (s)	lags used
10	10–30 (3 nearest)	90	60–140	320	210–470
20	20–40 (3 nearest)	100	60–170	390	260–580
30	20–50	110	80–170	470	320–710
40	30–60	140	80–210	580	390–1000
50	30–80	170	100–260	710	470–1000
60	40–100	210	140–320	870	580–1000
80	50–140	260	170–390	1000	580–1000

Table 6: Lags of the window of each H_{loc} value. At 10 and 20 s the window is not centred; at 870 and 1000 s it is cut at 1000 s and the two values use the same lags, so the end of the curve flattens by construction. The window resolves nothing narrower than about half a decade.

B Which statistics are drawn

Ratios. Ratios are sums of numerators over sums of denominators over the runs of a population (a run goes to the stretch that holds its middle). In a small stratum a glide in rising air has almost no height lost, the glide ratio diverges and its average over strata is unstable; stratified contrasts use the *height lost per km of glide*, the inverse. Pooled tables keep the glide ratio.

Redundancy. Unit: a population on a site-day with at least 25 flights (119 units). A statistic is redundant when the others explain most of it across units (R^2 of its linear regression on them); backward elimination drops the most explained until every $R^2 \leq 0.5$. The four shares of time sum to 100%, so one of them goes first.

statistic	status	R^2 on the five drawn
time in climb (%)	drawn	0.23
searches that end in a climb (%)	drawn	0.41
climb rate (m/s)	drawn	0.20
glide speed (km/h)	drawn	0.26
glide ratio	drawn	0.38
time in transition (%)	dropped, number 1, R^2 1.00 when dropped	0.83
height gained per climb (m)	dropped, number 2, R^2 0.98 when dropped	0.91
climbs that start after a search (%)	dropped, number 3, R^2 0.93 when dropped	0.77
cross-country speed over 30 min (km/h)	dropped, number 4, R^2 0.93 when dropped	0.90
time in a search in vain (%)	dropped, number 5, R^2 0.88 when dropped	0.71
climbs per hour	dropped, number 6, R^2 0.72 when dropped	0.71
time in a search that ends in a climb (%)	dropped, number 7, R^2 0.65 when dropped	0.65

Table 7: Five statistics remain; the others are explained by them.

C Robustness to the threshold

With $q = 0.2$ a contact needs only 26.4 min: groups are more and shorter, and up to one contact in five may be chance.

	$q = 0.1$, $\tau^* = 38.1$ min	$q = 0.2$, $\tau^* = 26.4$ min
group flights (TA-MSD)	33 435	48 777
$H_{\mathrm{loc}}(1000\ \mathrm{s})$, S	0.837	0.836
$H_{\mathrm{loc}}(1000\ \mathrm{s})$, A	0.860	0.854
$H_{\mathrm{loc}}(1000\ \mathrm{s})$, G	0.893	0.892
$\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$, G − A same flight	+0.037 [+0.035, +0.039]	+0.032 [+0.029, +0.035]
$\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$, G − A same s-d-h	+0.043 [+0.040, +0.046]	+0.042 [+0.040, +0.044]
$\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$, A − S same s-d-h	+0.003 [+0.001, +0.006]	+0.002 [−0.001, +0.005]
$\Delta H_{\mathrm{loc}}(100\ \mathrm{s})$, G − A same flight	−0.009 [−0.009, −0.008]	−0.016 [−0.017, −0.015]
$\Delta H_{\mathrm{loc}}(100\ \mathrm{s})$, G − A same s-d-h	+0.002 [+0.001, +0.002]	−0.004 [−0.005, −0.003]
Δ searches to climb, G − A same s-d-h (points)	+2.6 [+2.3, +3.0]	+3.4 [+3.0, +3.7]
Δ climb rate, G − A same s-d-h (m/s)	+0.029 [+0.025, +0.034]	+0.024 [+0.020, +0.029]
Δ height lost per km, G − A same s-d-h (m/km)	−9.5 [−10.5, −8.6]	−11.5 [−12.2, −10.9]
Δ cross-country speed, G − A same s-d-h (km/h)	+1.29 [+1.16, +1.41]	+1.14 [+1.01, +1.30]

Table 8: The main results with the two thresholds (s-d-h: same site-day and hour). Every result at 1000 s and every phase statistic keeps its sign; at 100 s the estimates turn slightly negative, consistent with no effect within one phase.

D Pooled curves, hang gliders and sizes

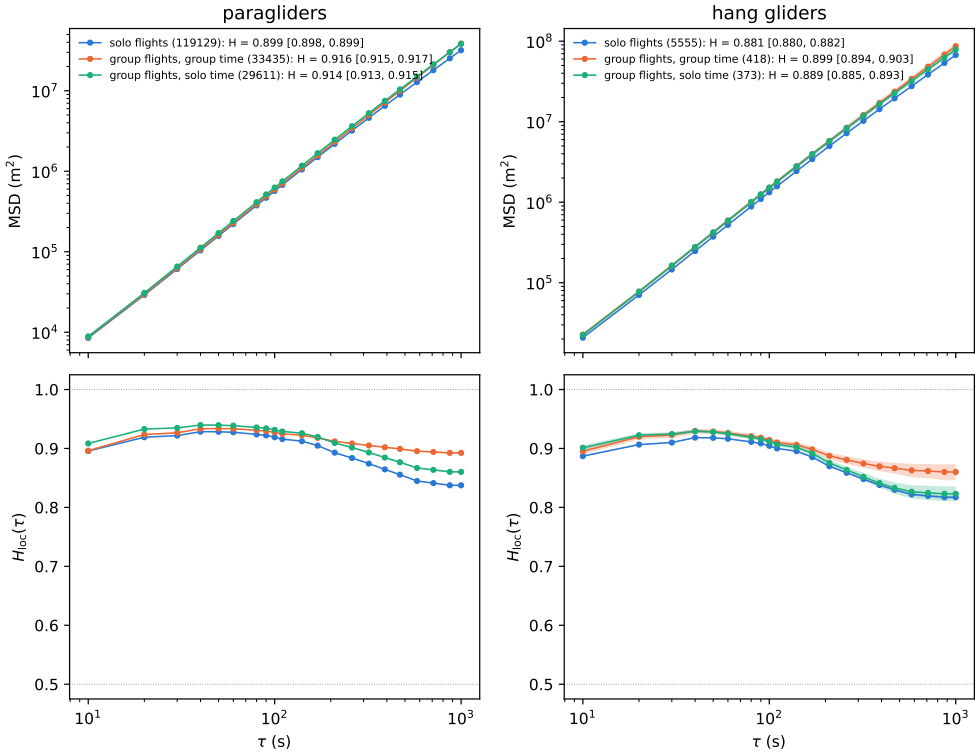


Figure 7: TA-MSD and H_{loc} of S, G and A, paragliders and hang gliders (hang gliders: 418 group flights, wide bands at long lags).

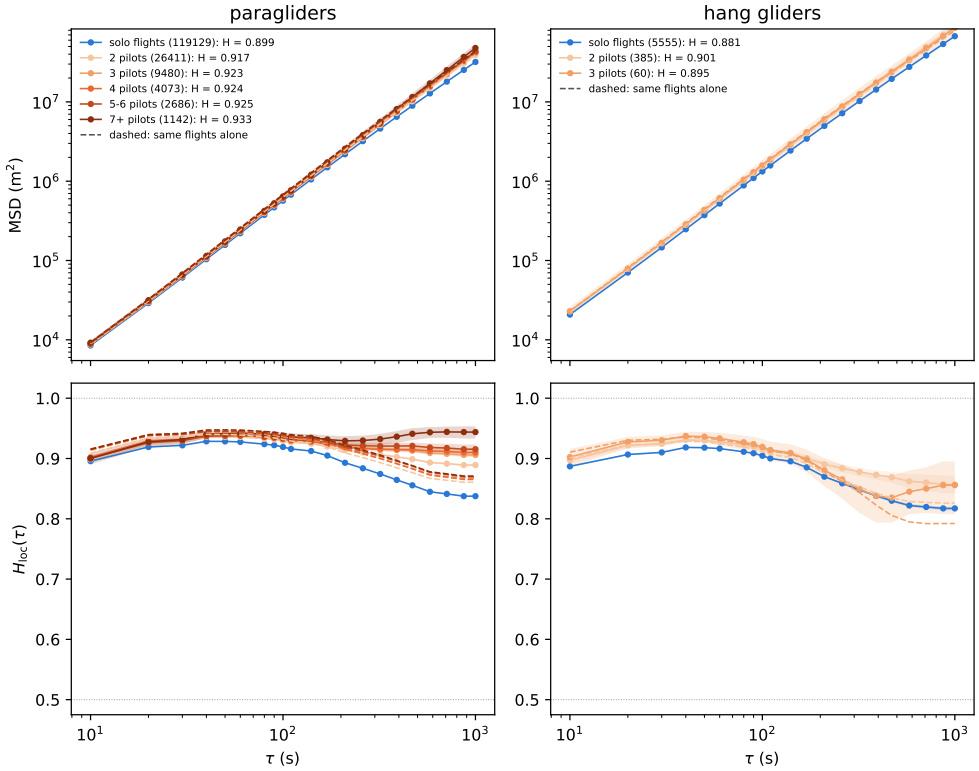


Figure 8: By the size of the configuration (orange, darker with more pilots), S (blue) and, dashed, the same flights alone for each size.

E Tables and figures of the tests

Δ	G − S pooled	G − S same s-d-h	A − S same s-d-h	G − A same flight	G − A same s-d-h
$H_{loc}(100\ s)$	+0.008 [+0.007, +0.009]	+0.002 [+0.001, +0.003]	+0.005 [+0.004, +0.005]	−0.009 [−0.009, −0.008]	+0.002 [+0.001, +0.002]
$H_{loc}(1000\ s)$	+0.055 [+0.053, +0.057]	+0.041 [+0.038, +0.044]	+0.003 [+0.001, +0.006]	+0.037 [+0.035, +0.039]	+0.043 [+0.040, +0.046]
time in climb (%)	+2.20 [+1.96, +2.45]	+3.43 [+3.19, +3.63]	+0.43 [+0.25, +0.59]	+5.71 [+5.52, +5.92]	+3.28 [+3.11, +3.47]
searches to climb (%)	+6.18 [+5.82, +6.46]	+4.12 [+3.68, +4.54]	+1.60 [+1.23, +1.97]	+8.75 [+8.37, +9.15]	+2.59 [+2.25, +2.99]
climb rate (m/s)	+0.040 [+0.030, +0.051]	+0.031 [+0.025, +0.039]	+0.007 [+0.001, +0.012]	+0.024 [+0.019, +0.031]	+0.029 [+0.025, +0.034]
glide speed (km/h)	+3.18 [+2.97, +3.39]	+1.97 [+1.82, +2.10]	+1.15 [+1.05, +1.27]	+1.36 [+1.27, +1.46]	+0.87 [+0.79, +0.98]
height lost per km (m/km)	−18.6 [−19.3, −17.8]	−10.0 [−10.7, −9.1]	−0.5 [−1.1, +0.3]	−21.1 [−22.1, −20.1]	−9.5 [−10.5, −8.6]
cross-country speed (km/h)	+2.81 [+2.59, +3.01]	+2.25 [+2.09, +2.39]	+1.31 [+1.21, +1.43]	+0.70 [+0.60, +0.78]	+1.29 [+1.16, +1.41]

Table 9: The five contrasts of Section 4 (Figure 1): difference and 5–95% band (s-d-h: same site-day and hour).

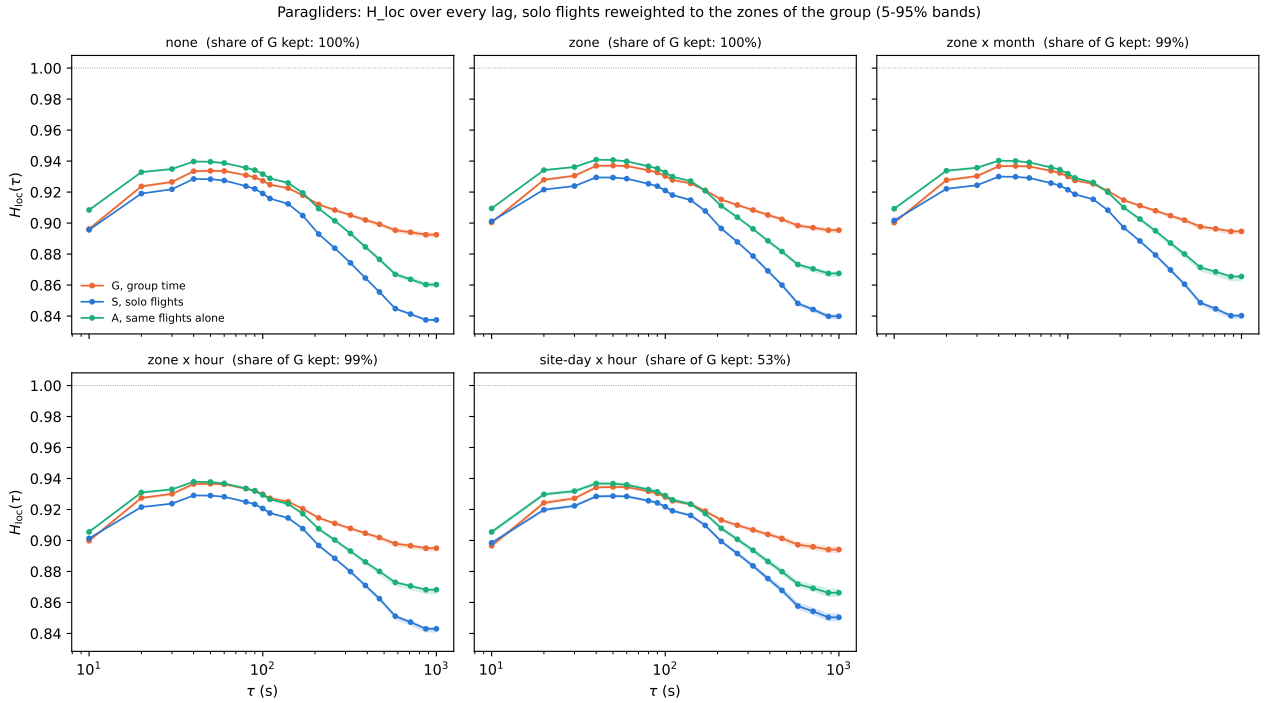


Figure 9: $H_{\mathrm{loc}}(\tau)$ of G (orange), of S reweighted to the strata of G (blue) and of A reweighted to them (green), 5–95% bands (Section 5). First panel: all flights pooled. "Share of G kept": flights of G with a solo counterpart in their stratum.

comparison	events	time in climb (points)	searches to climb (points)	climb rate (m/s)	glide speed (km/h)	height lost per km (m/km)
entering: start − before	4 786	+0.7 [+0.3, +1.1]	+0.0 [−1.1, +1.0]	+0.046 [+0.034, +0.062]	+0.92 [+0.69, +1.15]	−2.3 [−3.5, −1.3]
leaving: end − after	10 097	+1.5 [+1.2, +1.9]	+1.2 [+0.4, +1.9]	−0.007 [−0.017, +0.003]	+0.52 [+0.35, +0.64]	−10.9 [−11.8, −10.0]
after − before	4 718	−1.4 [−1.8, −1.0]	−2.8 [−4.0, −1.6]	+0.043 [+0.024, +0.059]	+0.33 [+0.12, +0.55]	+9.1 [+7.8, +10.4]
control A: later − earlier	12 568	−1.1 [−1.3, −0.9]	−1.4 [−2.0, −0.8]	+0.018 [+0.009, +0.027]	+0.07 [−0.05, +0.20]	+3.3 [+2.5, +4.2]
control S: later − earlier	49 092	−1.6 [−1.7, −1.5]	−2.3 [−2.7, −2.0]	+0.011 [+0.008, +0.015]	+0.04 [−0.03, +0.10]	+4.9 [+4.5, +5.3]

Table 10: The phase statistics of the comparisons of Section 6 (differences with 5–95% band); runs and glides are placed in the window that holds their middle. Leaving is end minus after: its drift is the control with the opposite sign.

	G_n − A, same flight					G_n − G_2, same site-day-hour			
n	2	3	4	5–6	7+	3	4	5–6	7+
$H_{\mathrm{loc}}(100\ \mathrm{s})$	−0.008	−0.014	−0.012	−0.021	−0.013	−0.003	−0.002	−0.013	+0.009
$H_{\mathrm{loc}}(1000\ \mathrm{s})$	+0.026	+0.027	+0.043	+0.038	+0.087	+0.004	−0.001	−0.015	+0.064
time in climb (%)	+5.76	+8.26	+9.37	+9.11	+9.91	+1.98	+3.77	+3.54	+4.03
searches ending in a climb (%)	+8.25	+9.41	+10.78	+9.97	+11.25	+0.28	+0.55	+0.35	+1.19
climb rate (m/s)	+0.021	+0.006	−0.008	−0.012	−0.055	+0.002	−0.003	−0.006	−0.017
glide speed (km/h)	+1.29	+1.15	+1.32	+0.93	+0.96	−0.23	−0.40	−0.46	−0.18
height lost per km of glide (m/km)	−17.6	−22.8	−28.1	−32.3	−28.1	−3.0	−8.0	−8.7	−7.5

Table 11: The contrasts by size of Section 7.1 (Figure 6).

F Before, during and after with short windows

Same test as Section 6 with windows of 21 min, the shortest a piece of the TA-MSD allows. Groups of at least 21 min enter: more events, but at 1000 s a window holds fewer than thirty origins. The step at 1000 s is then not seen (−0.003 entering, +0.008 leaving, against drifts of −0.011 and −0.007), and the fitted H, dominated by short lags, shows no change beyond the drift. The phase statistics at entry depend on the window (time in climb +3.3 points, glide speed +0.63 km/h here, against +0.7 and +0.92 with 50 min).

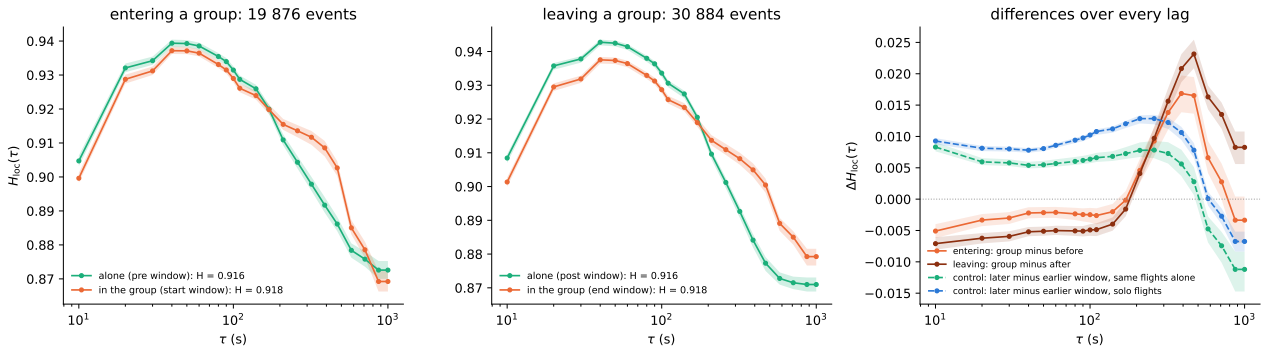


Figure 10: As Figure 3, windows of 21 min.

comparison	events	ΔH	$\Delta H_{\text{loc}}(100)$	$\Delta H_{\text{loc}}(1000)$
entering: start − before	19 876	+0.002 [+0.001, +0.004]	−0.002 [−0.004, −0.001]	−0.003 [−0.007, +0.000]
leaving: end − after	30 884	+0.002 [+0.001, +0.003]	−0.005 [−0.006, −0.004]	+0.008 [+0.006, +0.011]
after − before	14 482	+0.004 [+0.003, +0.005]	+0.006 [+0.005, +0.008]	+0.002 [−0.001, +0.006]
control A: later − earlier	33 658	+0.006 [+0.005, +0.006]	+0.006 [+0.006, +0.007]	−0.011 [−0.015, −0.008]
control S: later − earlier	107 324	+0.009 [+0.009, +0.010]	+0.010 [+0.010, +0.011]	−0.007 [−0.008, −0.005]

Table 12: As Table 3, windows of 21 min.

comparison	events	time in climb (points)	searches to climb (points)	climb rate (m/s)	glide speed (km/h)	height lost per km (m/km)
entering: start − before	14 798	+3.3 [+2.9, +3.8]	+1.2 [+0.3, +2.2]	+0.043 [+0.035, +0.053]	+0.63 [+0.45, +0.77]	−8.9 [−12.1, −6.2]
leaving: end − after	23 014	+2.7 [+2.3, +3.1]	+0.9 [+0.2, +1.6]	+0.008 [+0.000, +0.018]	+0.54 [+0.42, +0.67]	−13.1 [−14.5, −11.5]
after − before	10 898	−1.5 [−2.0, −1.1]	−4.1 [−5.2, −2.9]	+0.017 [−0.001, +0.030]	+0.17 [−0.01, +0.35]	+9.6 [+5.7, +12.9]
control A: later − earlier	25 121	−2.1 [−2.4, −1.8]	−2.8 [−3.5, −2.0]	−0.001 [−0.009, +0.006]	−0.29 [−0.40, −0.15]	+3.4 [+1.9, +5.2]
control S: later − earlier	75 984	−2.0 [−2.2, −1.8]	−3.0 [−3.5, −2.7]	−0.012 [−0.017, −0.009]	−0.31 [−0.37, −0.24]	+6.2 [+5.0, +7.5]

Table 13: As Table 10, windows of 21 min.

G Single site-days

Six site-days chosen by a fixed rule (most sizes with 15 flights or more, then most flights, at most two per takeoff site). Bands resample flights (too narrow: flights of one group are not independent).

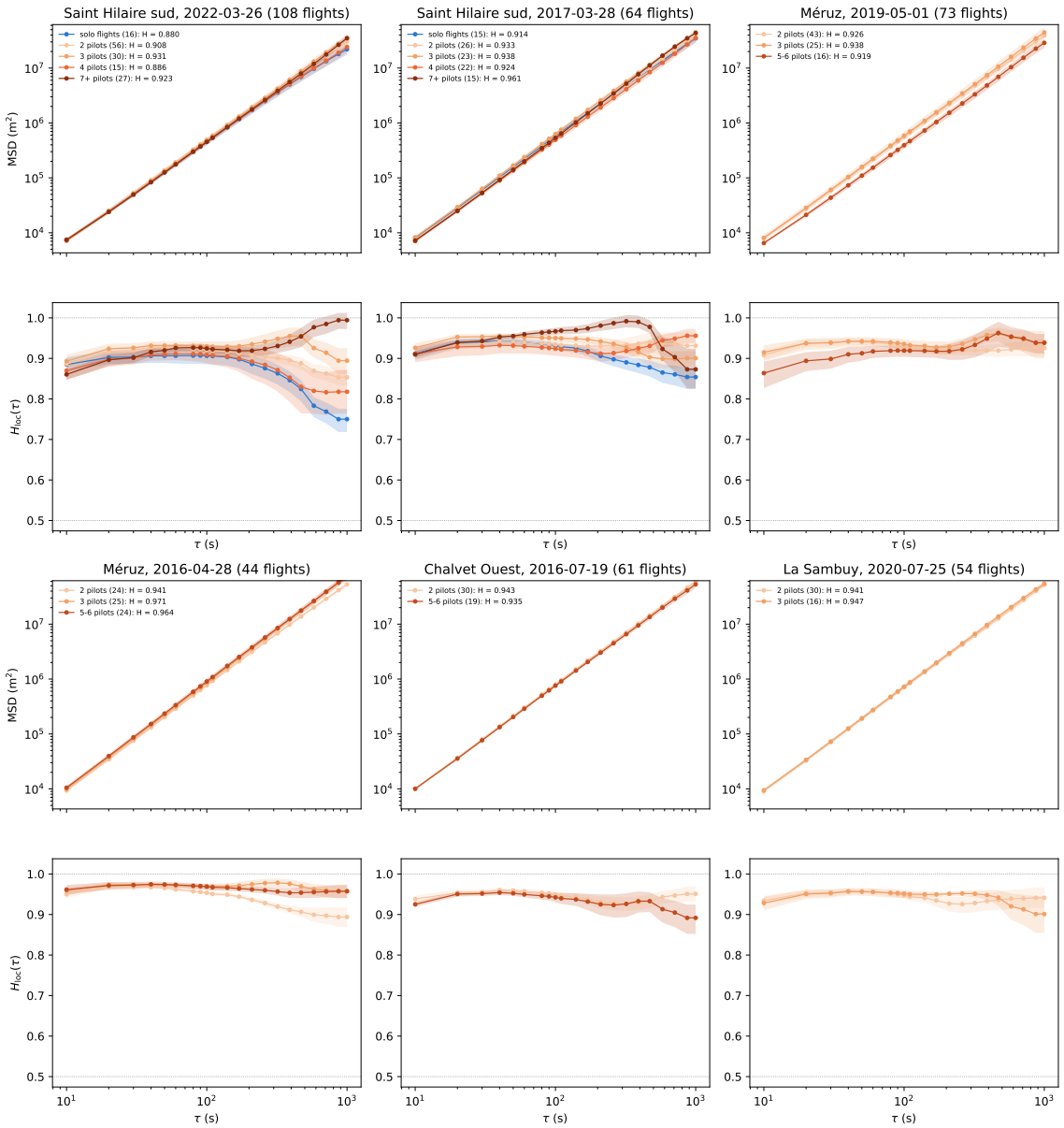


Figure 11: TA-MSD and H_{loc} on six site-days by the size of the configuration, with the solo flights of the same site-day where they reach 15 flights.

site-day	solo	2	3	4	5–6	7+
Saint Hilaire sud, 2022-03-26	0.750	0.853	0.894	0.818		0.994
Saint Hilaire sud, 2017-03-28	0.854	0.931	0.901	0.956		0.873
Méruz, 2019-05-01		0.926	0.939		0.938	
Méruz, 2016-04-28		0.894	0.958		0.957	
Chalvet Ouest, 2016-07-19		0.951			0.892	
La Sambuy, 2020-07-25		0.941	0.901			

Table 14: $H_{\text{loc}}(1000\text{ s})$ by size on each site-day (empty: fewer than 15 flights). Differences between days exceed those between sizes (the climb rate goes from 0.79 to 2.11 m/s across the six days).

H Leader minus follower

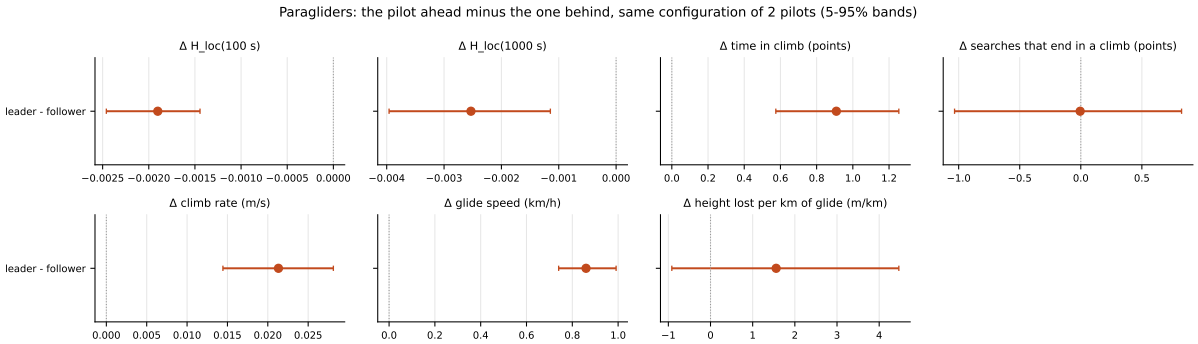


Figure 12: Leader minus follower, same configuration of two pilots.

Δ	leader − follower
$H_{\text{loc}}(100\ \text{s})$	-0.002 [-0.002, -0.001]
$H_{\text{loc}}(1000\ \text{s})$	-0.003 [-0.004, -0.001]
time in climb (%)	$+0.91$ [$+0.57$, $+1.25$]
searches ending in a climb (%)	-0.01 [-1.03, $+0.83$]
climb rate (m/s)	$+0.021$ [$+0.014$, $+0.028$]
glide speed (km/h)	$+0.86$ [$+0.74$, $+0.99$]
height lost per km of glide (m/km)	$+1.6$ [-0.9, $+4.5$]
cross-country speed (km/h)	$+0.14$ [$+0.11$, $+0.16$]

Table 15: Leader minus follower, same configuration of two pilots (difference with 5–95% band).

I Checks of the viewer and of the analyses

The checks of the grouping itself, its code against the note and the properties of its tables, are in section 8 of the note.

- **Viewer.** The *Group flights* tab reads `groups/` of its data folder: its seven tables are **byte-identical** to those of the chosen run; the tracks it draws (`SSD_DATA`) have the same fixes, times, positions and trimmed starts as those the grouping read (`SSD_DISANTE`).
- **Analyses.** Tests on synthetic flights for every step, and on every number of the README of the analyses; the decomposition is checked on a synthetic case where the day raises a statistic by 10 and the group by 2: pooled 10, stratified 2.

J Reproducing the report

From `xc-lab/analysis/groups/group-identification`, SSD_DISANTE mounted:

```
uv run python tamsd.py; uv run python phase_msd.py; uv run python survival.py
uv run python budget.py; uv run python performance.py; uv run python blocks.py
uv run python redundancy.py; uv run python size.py; uv run python decompose.py
uv run python zones.py; uv run python prepost.py; uv run python prepost.py --window 1260
uv run python mechanisms.py; uv run python report/facts.py
uv run python report/tables.py; cd report; latexmk -pdf group_vs_solo.tex
```

Heavy pieces stay on SSD_DISANTE, in the `analyses/` folder of each run; figures and JSON in `out/<run>/`.